

Extending Deep Generative Markov State Models: Non-Markovian and Continuous Dynamics

Project Description

Context

Many complex dynamical systems arising in physics, biology, and engineering exhibit multi-scale behavior, where fast fluctuations coexist with rare transitions between long-lived metastable states. A central objective is to construct reduced models that capture the essential long-time dynamics while remaining computationally tractable.

A recent approach, known as *Deep Generative Markov State Models*, combines representation learning and stochastic modeling to address this challenge. The method learns a low-dimensional latent representation of the system, models its evolution as a Markov process, and uses a generative component to reconstruct configurations and simulate trajectories. This framework enables the approximation of long-time kinetics and the generation of realistic samples from high-dimensional dynamical systems[4].

Despite its success, this approach relies on two fundamental assumptions. First, it assumes that the latent dynamics are Markovian at a chosen time discretization, i.e., the future state depends only on the present latent variable. Second, it assumes that the latent representation can be adequately described by a finite set of discrete metastable states.

These assumptions are often violated in practice. Coarse-graining procedures typically introduce memory effects, leading to non-Markovian dynamics at the latent level. Furthermore, many systems do not admit a natural partition into a small number of discrete states, but instead evolve on continuous low-dimensional manifolds. As a result, standard Markov state models may fail to capture important dynamical features, such as long-range temporal correlations or smooth transitions between metastable regions.

Goal

The objective of this project is to investigate extensions of deep generative Markov state models that relax these limitations. In particular, the goal is to

develop models that (i) account for non-Markovian effects in the latent dynamics, and (ii) replace discrete latent states with continuous representations. These extensions aim to provide a more faithful and flexible description of complex dynamical systems, bridging the gap between classical Markov state models and modern deep latent dynamical systems. The project will focus on Alanine dipeptide. Identifying an appropriate dataset and selecting suitable evaluation metrics are integral parts of the project.

References to use for project

- [1] Ricky TQ Chen, Yulia Rubanova, Jesse Bettencourt, and David K Duvenaud. Neural ordinary differential equations. *Advances in neural information processing systems*, 31, 2018.
- [2] Mathias Schreiner, Ole Winther, and Simon Olsson. Implicit transfer operator learning: Multiple time-resolution models for molecular dynamics. *Advances in Neural Information Processing Systems*, 36:36449–36462, 2023.
- [3] Yuning Shen, Lihao Wang, Huizhuo Yuan, Yan Wang, Bangji Yang, and Quanquan Gu. Simultaneous modeling of protein conformation and dynamics via autoregression. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [4] Hao Wu, Andreas Mardt, Luca Pasquali, and Frank Noé. Deep generative markov state models. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 3979–3988, 2018.